

Addis Ababa University
College of Health Sciences
Department of Physiology and Anatomy
Unit of Physiology

Physiology II final examination for Pharmacy regular year two students
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Academic year: 2013
Date of examination: January 25, 2013
Time allowed: 2:30 hours
Total mark obtained: _____
% of marks obtained: _____

Name of the student: _____

ID. No: _____

Exam profile

Items	Number	Each point	Total point
MCQs	60	1	60
True/false	10	0.5	5
Matching	24	0.5	12
Total	94		77

Note: check the presence of all 13 consecutive pages

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January 2013

Part I. Multiple choice questions (MCQs), 1 each, 1x60=60)

Directions: This format contains 60 numbered items. All the questions or incomplete statements are followed by the answer or completion of the statement. **Underline the letter of your choice**

1. Which of the following enzymes is found in the saliva
 - A. Ptyalin
 - B. Pepsin
 - C. Trypsin
 - D. Chemotrypsin
 - E. enterokinase
2. Which of the following food substance not need digestion
 - A. Vitamins
 - B. Carbohydrates
 - C. Proteins
 - D. Triglycerides
 - E. Maltose
3. Which of the following is not true about digestive system
 - A. All the food processes are regulated
 - B. Food digestion processes are initiated by extrinsic autonomic nerve
 - C. Preganglionic parasympathetic fibers enter into the stomach
 - D. The hormones in the GIT released into the duodenal wall
 - E. Stomach parietal cells released the HCl acid in its form
4. Which of the following processes is not the function of the digestive system
 - A. Breaking down of the food substances starting from mouth
 - B. Involving in the homeostasis of the body
 - C. Chemical digestion in the large intestine by human enzymes
 - D. Making the digested food substance available into the tissues
 - E. Producing of enzymes and hormones for the digestion processes
5. In which of the following areas where the kupffer cells that engulf and digest bacteria when they passed through together food
 - A. Hepatic portal artery
 - B. Hepatic portal vein
 - C. Cystic duct wall
 - D. Common bile duct
 - E. Pancreatic duct

6. Which of the following intestinal hormones facilitate the release of the bile salt from the gallbladder highly than the other
- | | |
|-----------------------------|--------------|
| A. Acetylcholine | D. Motiline |
| B. Gastroinhibitory peptide | E. Secretion |
| C. Cholecystokinin | |
7. Which of the following hormones is highly involved in the neutralization of the chime acid in the duodenum
- | | |
|-----------------------------|--------------|
| A. Gastrin | D. Motiline |
| B. Gastroinhibitory peptide | E. Secretion |
| C. Cholecystokinin | |
8. Which of the following hormones not reduces the stomach activities
- | | |
|-----------------------------|-----------------|
| A. Gastrin | D. noradrenalin |
| B. Gastroinhibitory peptide | E. Secretion |
| C. Cholecystokinin | |
9. Which of the following is true about the gastrointestinal tract and process in each tracts
- The secondary saliva in the mouth and blood has the same ionic concentration with the blood
 - Salivary gland Sympathetic stimulation increases the secretion of thick saliva
 - Nicotinic blockers in the salivary glands reduces the serous saliva secretion
 - Dehydration increases the saliva secretion from the salivary glands
 - Muscarnic stimulation in the salivary gland increases the saliva secretion
10. In which of the following gastrointestinal structures were the first peristalsis is started
- | | |
|----------|------------|
| A. Mouth | B. Stomach |
|----------|------------|

- C. Pharynx
- D. Esophagus

E. Stomach

11. Which of the following is true about the Gastroesophageal reflux disease

- A. Salivation and secondary peristalsis aggravates the diseases symptoms
- B. The tonic contraction of the lower esophageal sphincter is elevated in the patients
- C. Parietal cells histamine blockers reduce the symptoms manifested
- D. the symptoms are reduced during the pregnancy
- E. parietal cells nicotinic blockers increase the symptoms of the disease

12. Which of the following nutrients not move into the liver rather enter into the lymphatic vessels that is joined the thoracic duct

- A. Glucose
- B. Tripeptide
- C. Chylomicron
- D. Amino acids
- E. Fructose

13. Which of the following is not true

- A. Iron is stored in the liver in the form of ferritin
- B. Enterohepatic circulation is increased during meal
- C. Bile is concentrated during the meal
- D. Vitamin B12 needs carrier protein to be absorbed
- E. Digested fat is absorbed passively in the small intestine

14. Which of the following digested food substance absorbed into the intestinal tubular epithelial cell together with hydrogen ion in the apical membrane

- A. Dipeptide
- B. Glucose
- C. Long chain fatty acid
- D. Short chain fatty acid
- E. Amino acid

15. From which of the following gastrointestinal tracts where the enzyme that digest and Convert di- and tripeptides into amino acids is basically produced

- A. Stomach
- B. Pancreatic gland

- C. Liver
- D. Small intestine
- E. Large intestine

16. Which of the following chemical is involved in the formation of micelle

- A. Pancreatic lipase
- B. Lingual lipase
- C. Bile salt
- D. Bile pigment
- E. Enterokinase

17. From which of the following gastrointestinal tracts where the enzyme that convert trypsinogen into trypsin is produced

- A. Stomach
- B. Pancreatic gland
- C. Small intestine
- D. Large intestine
- E. Liver

18. Which of the following is the brush border enzymes that convert small fragmentized peptides into amino acids

- A. Amino peptidase
- B. Carboxypeptidase
- C. Chemotrypsinogen
- D. Pepsin
- E. Chemotrypsin

19. Which of the following is not true

- A. Fat emulsification occur in the duodenum
- B. Trypsin and chymotrypsin act in the small intestine
- C. Fat is not digested by the brush border enzymes
- D. Large amount of water absorbed in the large intestine
- E. No osmotic gradient that facilitate absorption of water in the stomach

20. Which of the following is an endocrine secretion

- A. Histamine
- B. Intrinsic factor
- C. Mucus
- D. Pepsinogen
- E. Trypsinogen

21. Which of the following is true

- A. Proteases in the pancreatic gland are active
- B. The non polar hydrophobic of the bile salt is face to the interior of the micelle
- C. Glucose enter into the intestinal absorptive cells by primary active transport

- D. The tonic contraction of the lower esophageal sphincter is reduced during acalachia
 - E. Stomach destination dose not stimulate the vomiting center in the medulla oblongata
22. Which of the following is not secreted from the stomach if the pyloric area of the stomach is removed
- A. Gastrin
 - B. Intrinsic factor
 - C. Hydrochloric acid
 - D. Pepsinogen
 - E. Mucus
23. All of the following reduces the gastric emptying process except one
- A. Duodenal hormones
 - B. Duodenal distension
 - C. Duodenal irritation
 - D. Duodenal emptying
 - E. Duodenal hormones
24. Which of the following enzymes is not effective in the small intestine
- A. Pepsin
 - B. Trypsin
 - C. Bile salt
 - D. Maltase
 - E. Carboxypeptidase
25. Which of the following is true about the gastric acid secretion processes
- A. Gastrin inhibits stomach acid secretion indirectly
 - B. Histamine from enterochromaffin like cells reduce acid secretion
 - C. Vagal stimulation increase the acid secretion in the stomach
 - D. Sympathetic blockers reduces acid secretion in the stomach
 - E. Potassium reach diet reduces acid secretion in the stomach
26. Observing the food substances bring changes in the body in such a way that the digestive enzyme secretion and activities in the stomach increased. This phase of gastric activity control is
- A. Intestinal phase
 - B. Duodenal phase
 - C. Gastric phase
 - D. Cephalic phase
 - E. Stomach phase
27. Which of the following is true about the lactose Intolerance

- A. Brush border lactase production is elevated during the disease
- B. Dehydration is the characteristic feature of the lactose intolerance
- C. Bacterial fermentation of carbohydrate in the large intestine is reduced
- D. Constipation is strongly manifested in the children having this intolerance
- E. Individuals weight is elevated due to the lactose intolerance

28. Select the one which is true about the digestive system

- A. Proteins and carbohydrates enter into the blood vessels nearer to the small intestine
- B. Hardening of the wall of the large intestine brings diverticulosis
- C. Most of the gastritis developed in the individuals are caused nonbacterial factors
- D. Descending colon of the large intestine is the site where excess amount of water is absorbed
- E. Breaking down of the large fat droplets into the smaller parts without enzyme occurs in the jejunum of the small intestine

29. All of the following are the target tissues for the hormones released from the posterior pituitary gland (oxytocine) except one

- | | |
|------------------|-----------|
| A. Blood vessels | D. Uterus |
| B. Kidney | E. Breast |
| C. Liver | |

30. Which of the following is not tropic hormone

- A. Aldosterone
- B. Follicle stimulating hormone
- C. Prolactin
- D. Thyroid stimulating hormone
- E. Adreno corticotropic hormone

31. Which of the following is more specific hormones than the other among the given below

- | | |
|-------------------|--------------------|
| A. Cortisol | D. Triiodothyronin |
| B. Growth hormone | E. Aldosterone |
| C. Adrenalin | |

32. Which of the following factors inhibit the production of growth hormones

- A. Excess glucose in the blood

- B. Less fatty acid in the blood
 - C. Less amino acid in the blood
 - D. Exercise
 - E. Stress
33. Which of the following is true about the hormone action
- A. Insulin facilitates ketone bodies production
 - B. Insulin inhibits amino acid exocytosis
 - C. Cortisol reduces protein synthesis in the liver
 - D. Insulin increase the fatty acid level in the blood
 - E. Hormone action mechanism is not depend on the hormone solubility
34. All of the following increase insulin secretion except one
- A. Sympathetic stimulation
 - B. Parasympathetic stimulation
 - C. Glucose dependent insulinotropic peptide
 - D. Excess Amino acid in the blood
 - E. Excess fatty acid in the blood
35. Which of the following processes is not developed in the body when the plasma glucose concentration is reduced
- A. Ketone bodies formation
 - B. Glycogenolysis
 - C. Gluconeogenesis
 - D. Glucagon production
 - E. Release of fatty acid from the tissue
36. Which of the following is not the effect of insulin
- A. Activates glycogen phosphorylase
 - B. Reduces protein catabolism
 - C. Increases protein synthesis
 - D. Increases glucose uptake by the brain cells
 - E. Increases fat synthesis
37. Which of the following is true
- A. The concentration of thyrotropin releasing hormone is less in hypothyroidism
 - B. Thyroid hormone concentration elevated in hypothyroidism
 - C. Thyroid stimulating hormone concentration is elevated in the hypothyroidism
 - D. Thyroid gland is under stimulated
 - E. Toxic goitre reduces body basal metabolic activities
38. Puffiness of the face and other body parts due to accumulation of hydrophilic polymers in those subjects having less thyroid hormone is
- | | |
|-------------------|---------------------|
| A. Exophthalmoses | D. Pheochromocytoma |
| B. Myxedema | E. Paget's disease |
| C. Cyanosis | |

39. Which of the following hormones are antagonistic to each other
- A. Growth hormone and insulin
 - B. Cortisol and glucagon
 - C. Norepinephrine and glucagon
 - D. Parathyroid hormone and Calcitriole
 - E. Aldosterone and atrial natriuretic peptide
40. Which of the following is not the effect of cortisol
- A. Hepatic cell protein catabolism
 - B. Fat cells lipid breaking down
 - C. Formation of glucose
 - D. Immune suppression
 - E. Prevent white cell migration
41. All of the following increase the growth hormone synthesis except one
- A. High extracellular amino acid
 - B. Somatostatin
 - C. Deep sleep
 - D. Hypoglycaemia
 - E. Low fat
42. Which of the following is false?
- A. Iodine deficiency brings mental retardation
 - B. Thyroid hormones reduces blood fatty acid level
 - C. Calcitonin inhibits osteoclasts cells in the bone
 - D. Parathyroid hormone reduces blood phosphate
 - E. Sympathetic stimulation facilitates aldosterone production
43. Which of the following is not the effect of prolactin?
- A. Mamogenesis
 - B. Milk production
 - C. Lactogenesis
 - D. Galactopoeisis
 - E. Milk letdown
44. Which of the following hormones induces response from the target tissue through the second messenger system?
- A. Aldosterone
 - B. Thyroxin
 - C. Progesterone
 - D. Noradrenalin
 - E. Cortisol

45. Which of the following hormones majorly can be given orally?
- | | |
|------------------------|--------------------------------|
| A. Insulin | D. Thyroid stimulating hormone |
| B. Parathyroid hormone | E. Adrenaline |
| C. Cortisol | |
46. Select the true idea given below
- Thyroid hormones are specific hormones
 - Steroid hormones are stored for future use
 - All amine hormones have receptor inside the cells
 - Noradrenalin taken as neurotransmitter
 - Tropic hormones are stored in the posterior pituitary glands
47. All of the following hormones facilitate the synthesis of new enzymes except one
- | | |
|--------------------|-----------------|
| A. Glucagon | D. Progesterone |
| B. Aldosterone | E. Androgen |
| C. Triiodothyronin | |
48. All of the followings are steroid hormone except one
- | | |
|---------------|----------------|
| A. Calcitonin | D. Cortisol |
| B. Androgen | E. Aldosterone |
| C. Estrogen | |
49. All of the following is the function of hormone carrier protein complex formation except one
- Increase the hormone life span
 - Inhibit excretion hormone
 - Transport hormone
 - Activate hormone
 - Reserve hormone in the blood
50. Which changes cannot be made on the target cells following hormone receptor complex formation?
- Changes on ion channel
 - Changes on enzymatic activities
 - Change on the gene expression
 - Change on the second messenger activities

- E. None of the above
51. Select the one which is true about the hormones
- A. Thyroid hormone binding globulin affinity to T_3 is greater than T_4
 - B. Thyroid hormone receptors are found inside the nucleus
 - C. Inhibition of the hypothalamic neurosecretary cells by T_3 is short loop negative feedback
 - D. Supraoptic nuclei in the hypothalamus release oxytocine
 - E. Nervous system does not regulate hormone synthesis
52. Which of the following factors does not stimulate hypothalamic cells to produce antidiuretic hormones?
- A. Hyperosmolality
 - B. Low blood pressure
 - C. Moistening of GIT
 - D. Low blood volume
 - E. Dehydration
53. Which of the following is true about the hypothalamus pituitary gland communication and pituitary cells and hormones?
- A. Oxytocine released from the posterior pituitary gland is following blood born hormone stimulation
 - B. Axons of the secretory cells release hypophysiotropic hormones in the median anterior pituitary gland cells
 - C. Antidiuretic hormone is synthesized in the posterior pituitary gland following arriving of action potential from the hypothalamus
 - D. Axons of the hypothalamic neurosecretary cells terminate in the neurohypophysis
 - E. Tropes are cells in the anterior pituitary gland and those are used only synthesis tropic hormones
54. Which of the following hormone abnormality brings buffalo torso chest
- A. Insulin
 - B. Glucagone
 - C. Cortisol
 - D. Androgen
 - E. Renine
55. Which of the following hormone is elevated in the blood in those subjects who are exposed for the chronic stress
- A. Corticotrope releasing hormone
 - B. Thyroprope releasing hormone

- C. Thyroid stimulating hormone
- D. Prolactin releasing hormone
- E. Melanocyte stimulating hormone

56. Which of the following is true about the adrenal gland hormones

- A. Aldosterone reduces the activities of the $\text{Na}^+\text{-K}^+$ pump at the basal membrane of the tubular epithelial cells
- B. Adrenal medullary hormones reduces the respiratory and the heart rate
- C. Adrenal medullary catecholamines induces Responses from the target cells slowly than catecholamines derived from the axon terminal of the sympathetic fibers
- D. Androgen from the zona reticularis of the adrenal cortex is high in concentration that brings male characteristics in the normal condition
- E. Cortisol from the zona fasciculata increase protein synthesis in the extrahepatic cells

57. There are hormones those mimic the action of the sympathetic fibers. Which of the following hormones is do you think induces sympathetic agonist effects in the target tissue

- | | |
|--------------|------------------------|
| A. Thyroxin | D. Growth hormone |
| B. Prolactin | E. Parathyroid hormone |
| C. Insulin | |

58. All of the following hormones are produced from the hypothalamic neurosecretary cells except one

- A. Growth hormone inhibiting hormone
- B. Oxytocine
- C. Estrogen
- D. L. Argenin vasopressin
- E. Thyropropin releasing hormone

59. Which of the following hormones has receptor on the surface of cells in the target tissue of the body

- | | |
|----------------------|----------------|
| A. Insulin | D. Aldosterone |
| B. Estrogen | E. Androgen |
| C. Tetraiodothyronin | |

60. Which of the following is not the characteristics of the catecholamines and peptide hormones

- A. Stored for the future use as the body need
- B. Use the second messenger system to induce responses
- C. Facilitate the synthesis of new enzymes
- D. Transported freely in the blood without carried on the carrier proteins
- E. Have short life spin relative to other hormone types

Format: I matching (0.5 each, 0.5x 24 = 12)

Direction: Match the items under column A with the appropriate conditions under column B and write the number for each item under column A from column B on the space provided under column A

1. Match the endocrine diseases disease under column A with the appropriate cases for each disease under column B

Column A

- _____ a) Cushing diseases
- _____ b) Con's disease
- _____ c) Pheocromocytoma
- _____ d)Osteomalasia
- _____ e) Cretinism
- _____ f) Exophthalmoses
- _____ g) Toxic goitre
- _____ h) Acromegaly
- _____ i) Diabetes mellitus
- _____ j) Adheson's disease

Column B

- 1. Chromaffin cells adenoma
- 2. Excess aldosterone secretion
- 3. Thyroid stimulating immunoglobulin
- 4. Thyroid hormone deficiency
- 5. Growth hormone excess
- 6. Insulin deficiency
- 7. Excesses parathyroid hormone
- 8. Thyroid hormone excess
- 9. Excess cortisol
- 10. Cortisol deficiency
- 11. Glucagon excess

2. Match the cells in the gastrointestinal tracts under column A and the secretion they secret under column B.

Column A

- _____ a) Enterochromaffin like cells
- _____ b) Oxyntic cells

Column B

- 1. Histamine
- 2. Intrinsic factors

- _____ c) G cells
- _____ d) Brush border cells
- _____ e) Duodenal J cells
- _____ f) Chief cells
- _____ g) Pancreatic ductal cells
- _____ h) Goblet cells
- _____ i) pancreatic acinar cells

- 3. Maltase
- 4. Gastrin
- 5. Bicarbonate ions
- 6. Pepsinogen
- 7. cholecystokinin
- 8. mucus
- 9. thrypsinogen
- 10. Secretin

3. Match the cells in the anterior pituitary gland under column A with the hormone they secrete under column B

Column A

- _____ a) Somatotropes
- _____ b) Lactotropes
- _____ c) Gonadotropes
- _____ d) Thyrotropes
- _____ e) Corticotropes

Column B

- 1. Luteinizing hormone
- 2. Prolactin
- 3. Adrenocorticotrophic hormone
- 4. Thyroid stimulating hormone
- 5. Growth hormone
- 6. Antidiuretic hormone
- 7. Melanocytes

Format: III true or false (0.5 each, 0.5x 10 = 5)

Direction: write true if the statement is correct or false if the statement is incorrect on the space provided at the end of each sentence.

1. Myenteric plexus controls the amount of local blood flow into the GIT at a given time. _____
2. Endocrine hormones control hypothalamus hormone secretion in the short loop feedback. _____
3. Hyperthyroidism increases heat production. _____
4. Hypothalamic hypophyseal portal vein takes hypothalamic hormones into the posterior pituitary gland. _____
5. Growth hormone increases blood fatty acid level for the source of energy. _____
6. Thyroid hormones are stored relatively for a long period of time. _____
7. Triiodothyronine up-regulates thyrotropin releasing hormone receptors on the thyrotropes in the anterior pituitary gland. _____
8. During fasting amino acid and fatty acid are released from adipose tissue cells. _____
9. Nipple suckling by the baby reduces dopamine production. _____
10. Glucose is highly absorbed in the jejunum through primary active transport on the apical membrane of the cells lining the GIT. _____